

Introduction to Cryptography and Security

Exercise 12: Hash functions

December 4, 2025

Let $H : M \rightarrow T$ be a collision resistant hash function. Which of the following is collision resistant: (as usual, we use $\parallel$ to denote string concatenation).

- ① $H'(m) = H(m \parallel m)$
- ② $H'(m) = H(m) \oplus H(m)$
- ③ $H'(m) = H(|m|)$ (i.e. hash the length of m)
- ④ $H'(m) = H(m) \oplus H(m \oplus 1^{|m|})$ (where $m \oplus 1^{|m|}$ is the complement of m)
- ⑤ $H'(m) = H(m) \parallel H(m)$
- ⑥ $H'(m) = H(0)$
- ⑦ $H'(m) = H(m) \parallel H(0)$

You are asked to find collisions on two compression functions:

$$f_1(x, y) = \text{AES}(y, x) \oplus y \quad \text{and}$$

$$f_2(x, y) = \text{AES}(x, x) \oplus y$$

where $\text{AES}(x, y)$ is the AES-128 encryption of y under key x .
Your goal is to find four distinct pairs

$$(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$$

such that

$$f_1(x_1, y_1) = f_1(x_2, y_2) \text{ and } f_2(x_3, y_3) = f_2(x_4, y_4).$$

In other words, the first two pairs are a collision for f_1 and the last two pairs are a collision for f_2 .

- We consider three different hash functions which produce outputs of lengths 64, 128 and 160 bits.
- After how many random inputs do we have a probability of $\lambda = 0.5$ for a collision?
- After how many random inputs do we have a probability of $\lambda = 0.1$ for a collision?

- Describe how exactly you would perform a collision search to find a pair x_1, x_2 such that $h(x_1) = h(x_2)$ for a given hash function h .
- What is the time complexity of this search?
- What are the memory requirements for this type of search if the hash function has an output length of n bits?

Passwords are stored in a database as hash values instead of plaintext. During login, the entered password is hashed and compared with the stored hash.

- 1 You obtained a hashed password database. Which attacks below can help recover passwords? What would be the impact of each?

Attack A: Break the one-way property of h .

Attack B: Find second preimages for h .

Attack C: Find collisions for h .

- 2 Why is a random *salt* usually added to passwords before hashing and stored together with the hash in the database?
- 3 Is an 80-bit hash function sufficiently secure for this application?